

📌 Notes by Amit Mahata

1/3

What is Artificial Intelligence?

AI is a science of making machines perform tasks that normally require human intelligence.

1950 – Can machine think?

Alan Turing → The Turing test

1955 – John McCarthy (Artificial Intelligence)

Researchers came together with an ambitious belief →

“Every aspect of learning and intelligence could, in principle, be described precisely enough for a machine to simulate.”

★ After 1955 AI winter was there for a long period.

1986 → Synthetic Intelligence

1997 → Deep Blue defeated Garry Kasparov.

IBM developed

⇓ Best chess player that time

Have machines become smarter than human?

Part 2: The Deep Learning & Generative AI Era

Following the early milestones, AI evolved from rules-based engines to statistical pattern recognition and large language models.

2006 - Deep Learning

Neural Net Revival

Geoffrey Hinton & team popularized Deep Belief Networks & Backpropagation training for deep architectures.

2012 - AlexNet

GPU Computer Vision

Alex Krizhevsky used NVIDIA GPUs for ImageNet competition — reducing error rates from 26% to 15%.

2017 - Transformers

Attention Is All You Need

Google introduced the Transformer architecture, replacing RNNs/LSTMs with Self-Attention mechanisms.

2022+ - Generative AI

LLMs & Diffusion

ChatGPT, Claude, Gemini & Midjourney bring AGI-level natural language reasoning and creative generation to millions.

“Scaling Laws showed that adding parameters, data, and compute predictably increases model reasoning capability.”

Analysis: Are Machines Smarter Than Humans?

The answer depends on whether we measure **Narrow Specialization** vs **General Intelligence (AGI)**.

Dimension	Human Brain 	AI Systems 
Processing Speed	~100 Hz signal transmission	GigaFLOPS / TeraFLOPS parallel compute
Memory Capacity	Selective, associative memory	Petabytes of lossless retrieval
Specialized Tasks (Chess/Math/Code)	Human champions surpassed	Superhuman performance (DeepBlue, AlphaGo)
Generalization & Empathy	Unmatched emotional, physical & common-sense reasoning	Hallucinations, lacks true biological experience

Conclusion: Machines excel at specialized computation & scale; humans excel at consciousness, empathy & novel reasoning!